MPG No. 13.3.46 $4'm'm$ (4'm'm setting) [Type III, tetragonal] [T tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & T_{xy} & 0 \\ T_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$T_{xy} = \mathbb{T}^{(1)}_{xy}$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzx} & 0 & 0 \\ 0 & T_{yzx} & 0 \\ 0 & 0 & T_{yzx} \end{bmatrix}$$

$T_{yzx} = \mathbb{T}^{(1)}_{xyz}$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzx} & 0 & 0 \\ 0 & T_{yzx} & 0 \\ 0 & 0 & -2T_{yzx} \end{bmatrix}$$

$T_{yzx} = \mathbb{M}^{(1)}_{v}$
$2T_{yzx} = 2\mathbb{M}^{(1)}_{v}$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & 0 & 0 \\ 0 & -T_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$T_{yzx} = \mathbb{M}^{(2)}_{v}$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & T_{yzzx} & 0 & 0 \\ T_{xxxy} & T_{xxxy} & T_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$T_{xxxy} = -\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{xy}$
$T_{zzxy} = 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(1)}_{xy}$
$T_{yzzx} = 2\mathbb{T}^{\beta(1)}_{4z} + \mathbb{T}^{(2)}_{xy}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -T_{xxxy} & -T_{xxxy} & -T_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)}$$
$$T_{zzxy} = -4\mathbb{M}_z^{\beta(1)} + \mathbb{T}_{xy}^{(3)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & T_{yzzx} & 0 \\ T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = 2\mathbb{T}_{xy}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxxy} \\ 0 & 0 & -T_{xxxy} \\ 0 & 0 & 0 \\ 0 & T_{yzzx} & 0 \\ -T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = \mathbb{M}_z^{\beta(2)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{yzzx} = -\mathbb{M}_z^{\beta(2)} - \mathbb{T}_{xy}^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & 0 \\ 0 & 0 & 0 & -T_{yzzx} & 0 & 0 \\ T_{xyxx} & -T_{xyxx} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = \mathbb{M}_z^{\beta(3)} + \mathbb{T}_{xy}^{(6)}$$
$$T_{xyxx} = \mathbb{M}_z^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{xxxy} & 0 & 0 & T_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & T_{xxyx} & 0 & 0 & T_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & T_{zzxy} & 0 & 0 & T_{zzxy} \\ 0 & 0 & 0 & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & T_{yzxz} & 0 & 0 & T_{yzzx} & 0 & 0 \\ T_{xyxx} & T_{xyxx} & T_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxy} = 2\mathbb{M}_z^{\beta(1)} + \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{xxyx} = 2\mathbb{M}_z^{\beta(1)} - \mathbb{M}_z^{\beta(2)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$
$$T_{zzxy} = -4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$
$$T_{yzzx} = -\mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$
$$T_{yzxz} = \mathbb{M}_z^{\beta(2)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(5)}$$
$$T_{xyxx} = -2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$
$$T_{xyzz} = 4\mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} - \mathbb{T}_{xy}^{(3)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{yzzx} & 0 & 0 & T_{yzxz} & 0 \\ 0 & 0 & 0 & -T_{yzxz} & 0 & 0 & -T_{yzzx} & 0 & 0 \\ T_{xyxx} & -T_{xyxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzzx} = \mathbb{M}_{z}^{\beta(3)} + 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)}$$
$$T_{yzxz} = \mathbb{M}_{z}^{\beta(3)} - 2\mathbb{T}_{xy}^{(4)} + \mathbb{T}_{xy}^{(6)}$$
$$T_{xyxx} = \mathbb{M}_{z}^{\beta(3)} - 2\mathbb{T}_{xy}^{(6)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & T_{xxxy} & 0 \\ T_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & T_{zzxy} & 0 \\ T_{xxyx} & 0 & 0 \\ T_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & T_{xyyy} & 0 \\ 0 & 0 & -\frac{T_{xxxy}}{2} - \frac{T_{xxyx}}{2} + T_{xyyy} + T_{zzxy} \end{bmatrix}$$

$$T_{xxxy} = -\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} - 2\mathbb{T}_{xy}^{(5)}$$
$$T_{zzxy} = 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + \mathbb{T}_{xy}^{(3)}$$
$$T_{xxyx} = -\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} + \mathbb{T}_{xy}^{(3)} + 2\mathbb{T}_{xy}^{(5)}$$
$$T_{yzzx} = 2\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(5)}$$
$$T_{xyyy} = -\mathbb{T}_{4z}^{\beta(1)} + \mathbb{T}_{xy}^{(1)} + 2\mathbb{T}_{xy}^{(2)} - \mathbb{T}_{xy}^{(3)}$$